Troubleshooting DNS clients

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

This article discusses how to troubleshoot issues from DNS clients.

Check IP configuration

1. Open a Command Prompt window as an administrator on the client computer.
2. Run the following command:

   Windows Command Prompt

   ```
   ipconfig /all
   ```

3. Verify that the client has a valid IP address, subnet mask, and default gateway for the network to which it is attached and being used.
4. Check the DNS servers that are listed in the output, and verify that the IP addresses listed are correct.
5. Check the connection-specific DNS suffix in the output and verify that it is correct.

If the client does not have a valid TCP/IP configuration, use one of the following methods:

- For dynamically configured clients, use the `ipconfig /renew` command to manually force the client to renew its IP address configuration with the DHCP server.
- For statically configured clients, modify the client TCP/IP properties to use valid configuration settings or complete its DNS configuration for the network.

Check network connection

Ping test

Verify that the client can contact a preferred (or alternate) DNS server by pinging the preferred DNS server by its IP address.

For example, if the client uses a preferred DNS server of 10.0.0.1, run this command at a command prompt:

Windows Command Prompt

```
ping 10.0.0.1
```

If no configured DNS server responds to a direct pinging of its IP address, this indicates that the source of the problem is more likely network connectivity between the client and the DNS servers. If this is the case, follow basic TCP/IP network troubleshooting steps to fix the problem. Keep in mind that ICMP traffic must be allowed through the firewall in order for the ping command to work.

DNS query tests

If the DNS client can ping the DNS server computer, try to use the following `nslookup` commands to test whether the server can respond to DNS clients. Because nslookup doesn't use the client's DNS cache, name resolution will use the client's configured DNS server.

Test a client

Windows Command Prompt

```
nslookup <client>
```

For example, if the client computer is named **client1**, run this command:

Windows Command Prompt

```
nslookup client1
```

If a successful response is not returned, try to run the following command:

Windows Command Prompt

```
nslookup <fqdn of client>
```

For example, if the FQDN is **client1.corp.contoso.com**, run this command:

Windows Command Prompt

```
nslookup client1.corp.contoso.com.
```

> **Note**
>
> You must include the trailing period when you run this test.

If Windows successfully finds the FQDN but cannot find the short name, check the DNS Suffix configuration on the DNS tab of the Advanced TCP/IP Settings of the NIC. For more information, see Configuring DNS Resolution.

Test the DNS server

Windows Command Prompt

```
nslookup <DNS Server>
```

For example, if the DNS server is named DC1, run this command:

Windows Command Prompt

```
nslookup dc1
```

If the previous tests were successful, this test should also be successful. If this test is not successful, verify the connectivity to the DNS server.

Test the failing record

Windows Command Prompt

```
nslookup <failed internal record>
```

For example, if the failing record was **app1.corp.contoso.com**, run this command:

Windows Command Prompt

```
nslookup app1.corp.contoso.com
```

Test a public Internet address

Windows Command Prompt

```
nslookup <external name>
```

For example:

Windows Command Prompt

```
nslookup bing.com
```

If all four of these tests were successful, run `ipconfig /displaydns` and check the output for the name that failed. If you see "Name does not exist" under the failing name, a negative response was returned from a DNS server and was cached on the client.

To resolve the issue, clear the cache by running `ipconfig /flushdns`.

Next step

If name resolution is still failing, go to the Troubleshooting DNS Servers section.

Last updated on 08/07/2020